



INBIOSIS WORKSHOP SERIES 2025

Transcriptomics: Functional Insights Made Easy

This workshop will **focus** on topics ranging from **basic experimental design** to **advanced downstream analyses** of **RNA-seq** data as well as **tips to publish in high-impact journals**.

Upon the completion of this **3-day hybrid hands-on workshop**, participants will be **capable** of the following:

- **Conceptualising** a transcriptome **experiment** with different **considerations**
- **Understanding** a complete **workflow** of RNA-seq data acquisition, processing, assembly, functional annotation, analysis, and interpretation
- **Aware** of the different analysis workflows and **software** packages in RNA-seq analysis
- **Exposure** to different approaches for **integrating** transcriptomics with other **omics** towards **systems understanding**
- **Network opportunity** with other researchers in the field & **sharing of knowledge** via voluntary **flash talks**
- Hands-on in **de novo transcriptomic analysis** and **mining** data
- Hands-on in downstream **functional analyses**: WEGO, KEGG pathway, clustering, functional enrichment

Why should you attend

- ✓ You're a **researcher, academician**, or **student** interested in transcriptomics and its applications.
- ✓ You're interested in **advanced skills** in RNA-seq analysis.
- ✓ You want **user-friendly / web-based platforms** for the interpretation of RNA-seq data.
- ✓ You got the transcriptome **results** from the sequencing vendor but not sure how to **interpret** them.
- ✓ You wish to **analyse** the transcriptome data **hands-on**, especially as a biologist instead of a data scientist.
- ✓ You want to learn on how to mine for **biological knowledge** from transcriptome data **without programming/coding**.
- ✓ You work on a transcriptomic project and want to learn on how to **publish** your results as a **high-impact** journal article.

This workshop is opened to **local & international** participants, from **beginner** to **advanced researchers** in **molecular biology / omics**. The **content** of this workshop is also **transferable** to other omics studies, such as **proteomics**.

17-19 JUN 2025

[0900-1700 GMT+8]

VENUE: HYBRID

TRAINER

Assoc. Prof. Dr. Goh Hoe Han

FEE

Online

MYR300

USD80 (International)

On-site INBIOSIS

MYR750

Register Now!

*Priority will be given to those who made payment before **10th Jun 2025**

Organised by



Further details

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TENTATIVE PROGRAMME

Day 1

Time	Activity/Event
0830	Registration
0900	Introduction
	Lecture 1: Principles of transcriptomics
1030	<i>Morning Break</i>
1045	Lecture 2: A crash course on RNA-seq analysis
1230	<i>Technical talk 1 – Current developments of NGS</i>
1300	<i>Lunch Break</i>
1400	Practical 1: Data pre-processing/exploration
1600	<i>Afternoon Break</i>
1615	<i>Participant flash talks</i> , Summary, discussion & wrap-up
1700	<i>End of Day 1</i>

Day 2

Time	Activity/Event
0900	Introduction to Day 2 & Recap
0910	Lecture 3: Exploring RNA-seq analysis for biological discovery (Functional Annotation, DEG & Downstream Analyses)
1030	<i>Morning Break</i>
1045	Practical 2: DEG analysis
1230	<i>Technical talk 2 – Advanced RNA-seq</i>
1300	<i>Lunch Break</i>
1400	Practical 3: Advanced statistical analysis: Multiple testing correction and FDR
1600	<i>Afternoon Break</i>
1615	<i>Participant flash talks</i> , Summary, discussion & wrap-up
1700	<i>End of Day 2</i>

Day 3

Time	Activity/Event
0900	Introduction to Day 3 & Recap
0910	Lecture 4: RNA-seq downstream analyses
1030	<i>Morning Break</i>
1045	Practical 4: Cluster analysis and visualisation
1230	<i>Technical talk 3 – Applications of RNA-seq</i>
1300	<i>Lunch Break</i>
1400	Practical 5: Functional enrichment analysis
1600	<i>Afternoon Break</i>
1615	Summary, discussion & conclusion
1700	<i>End of Workshop</i>

LOCATION MAP

